Characterizing Rolling-Shutter Distortions in a 1U CubeSat Camera Using LED Strobe Imaging

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Introduction

A rolling shutter sensor, the ArduCAM OV5642, was selected for ALEASAT due to its lower cost, reduced power consumption, and lower heat generation compared to global shutter sensors. Although rolling shutter readout can introduce image skew due to the relative motion between the CubeSat and Earth, these sensors provide sufficient resolution and sensitivity within mission constraints. The feasibility of deskewing, therefore, depends on the stability and predictability of the rolling shutter readout. Notably, camera testing is not addressed in Table I of ISO 19683 (Space Systems – Design qualification and acceptance tests of small spacecraft and units), motivating this work to characterize rolling shutter behaviour and address this gap experimentally.

Objective

The objective of this work is to develop a low-cost test fixture to characterize the rolling shutter readout of the ArduCAM OV5642 sensor and to use these results to assess the feasibility of developing deskewing algorithms. Specifically, this study investigates:

- ***Does the rolling shutter progress linearly on a row-by-row basis?***
- ***What is the rolling shutter readout time (time required to read through all pixel rows)?***
- ***Is the rolling shutter readout time consistent throughout different exposures?***

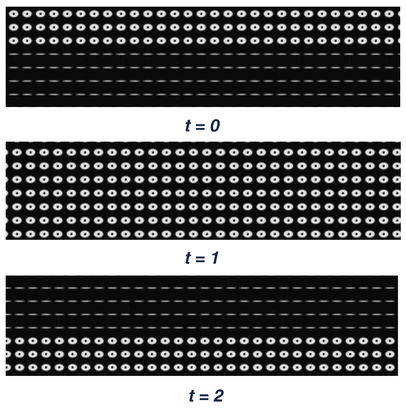
Background

What is Rolling Shutter?
Imagine each pixel as a tiny eye.
Each eye:

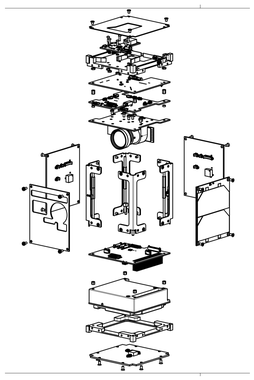
- Opens → Collects light → Closes

Unlike global shutters, pixels in a rolling shutter do not all capture at the same time.
Pixels open row by row, from top to bottom, creating a rolling shutter.

- **Readout time:** Time for all pixel rows to be read from top to bottom
- **Exposure time:** Time each pixel remains open to collect light



About ALEASAT



ALEASAT is a 1U (10 × 10 × 10 cm) Earth-observation CubeSat developed by UBC Orbit and SFU Satellite Design Team. The mission enables radio amateurs to request and directly downlink targeted Earth imagery, with applications in disaster monitoring and relief. Supported by the University of British Columbia, Radio Amateurs of Canada, and the Fly Your Satellite! Programme (ESA), ALEASAT is scheduled for launch in 2027.

The Payload Subteam develops, tests, and optimizes ALEASAT's camera system for Earth-observation missions. This includes firmware development and testing, optical analysis, image performance simulation, and in-orbit performance validation to ensure reliable and precise image capture.

Methodology

1. Use an LED strobe to periodically illuminate the sensor with pulses of fixed duration.
2. The rolling shutter exposes image rows sequentially during readout.
3. Periodic illumination produces horizontal intensity bands in the captured images.
4. Use the spacing between adjacent bands to determine rolling shutter linearity and estimate the total readout time.

Test Setup

- **Exposure times** are varied from **0.5 to 4 ms**
- **LED pulse durations** are varied from **2 to 16 ms**
- Measurements are performed separately for the **red, green, and blue** LED channels
- For each combination of exposure time, pulse duration, and LED wavelength, **5 to 10 images** are acquired
- A total of **150 images** are analyzed. More than **3000 intensity peaks** are identified across different exposure times, pulse durations, and LED wavelengths

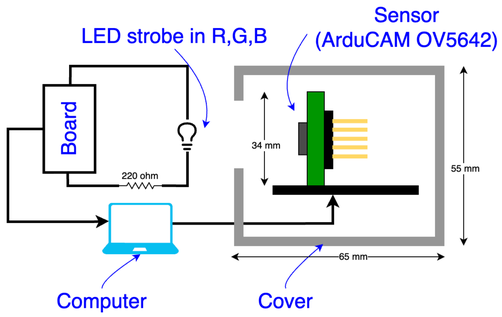


Fig 1. Test Setup Diagram

Results

Pulse Duration Dependence of Mean Interval at Different Exposures

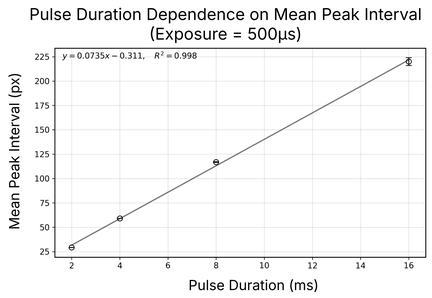


Fig 2a. Mean peak interval vs pulse duration (500 µs exposure)

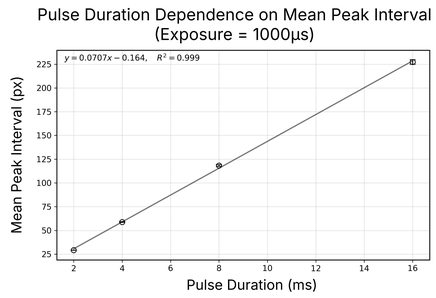


Fig 2b. Mean peak interval vs pulse duration (1000 µs exposure)

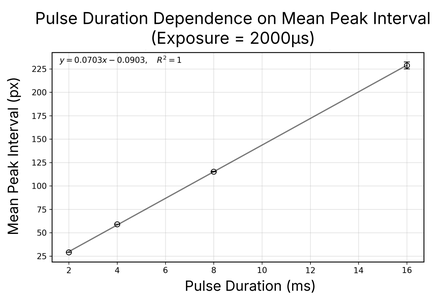


Fig 2c. Mean peak interval vs pulse duration (2000 µs exposure)

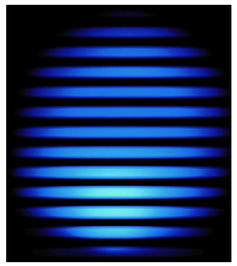


Fig 3. Sample output image

- **Mean peak interval increases linearly with pulse duration at all tested exposures** → rolling shutter "rows" in a linear fashion
- Linear fits at 500 µs, 1000 µs, and 2000 µs show near-unity R^2 values
- Pulse duration is the dominant factor governing peak interval scaling → **sensor readout is predictable & stable**

Readout Time Estimation

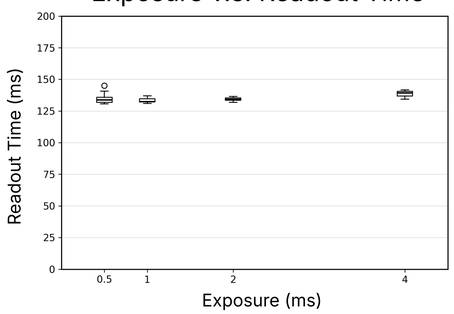


Fig 4. Exposure vs Readout Time

The readout time is estimated based on the data and the following formulae:

$$px\ per\ ms = \frac{mean\ peak\ distance\ (px)}{pulse\ duration\ (ms)}$$

$$readout\ time = \frac{n_{rows}}{px\ per\ ms}$$

- Sensor demonstrates a relatively consistent readout time at various exposures
- ***Readout time ~ 135 ± 3.18 ms***

Conclusion & Outlook

- The rolling shutter readout exhibits **linear, stable, and predictable behaviour** across all tested exposures
- The measured rolling shutter readout time is **135±3.18 ms**
- These results support rolling shutter image simulation and the development of deskewing algorithms
- The proposed method may inform future updates to ISO 19683

Acknowledgement

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